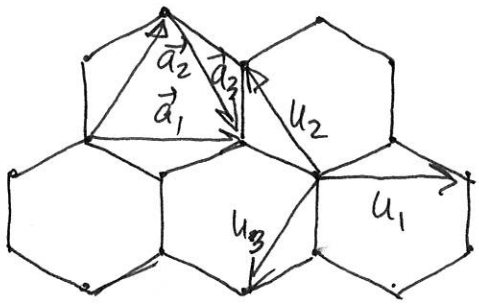
Top. insulator in graphene

$\vec{a}_{1,2,3}$ are vectors connecting nearest sites of the same sublattice. (p. 76)

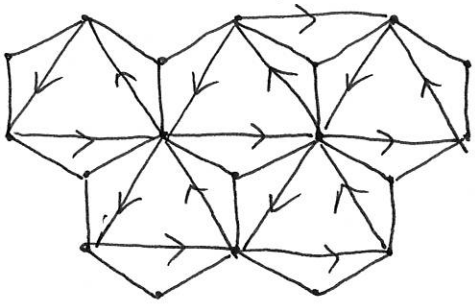
$\vec{a}_2 + \vec{a}_3 = \vec{a}_1$



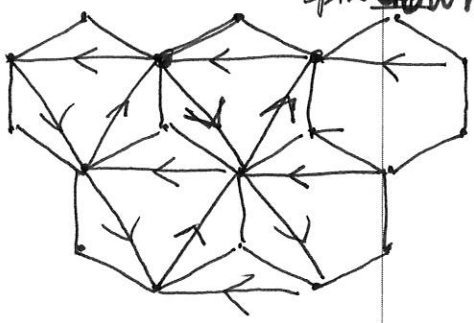
$$\hat{u}_1 = a\hat{x} = \vec{a}_1$$
$$\hat{u}_2 = a\left(-\tfrac{1}{2}\hat{x} + \tfrac{\sqrt{3}}{2}\hat{y}\right) = -\vec{a}_3$$
$$\hat{u}_3 = a\left(-\tfrac{1}{2}\hat{x} - \tfrac{\sqrt{3}}{2}\hat{y}\right) = -\vec{a}_2$$

(Kane, Mele 2005)

A sublattice spin <u>up</u>



B sublattice spin <u>down</u>



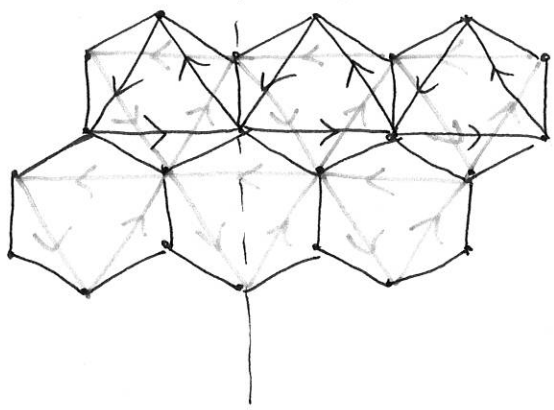
This is example of <u>spin-orbit coupling</u>.

$$H_{SO} = \lambda \sum_{\vec{r}\in A} \sum_{j=1}^{3} \left(i\, c^{\dagger}_{A\,\vec{r}+\vec{u}_j}\, S^z\, c_{A\,\vec{r}} + h.c. \right)$$
$$- \sum_{r\in B} \sum_{j=1}^{3} \left(i\, c^{\dagger}_{B\,\vec{r}+\hat{u}_j}\, S^z\, c_{B\,\vec{r}} + h.c. \right)$$

$\hat{S}^z$ = Pauli matrix acting on spin <u>indices</u>

$\hat{S}^z$ here means that spin $\uparrow$ hops with $(+i)$ on A sublattice along the arrows, and $\downarrow$ gets phase $(-i)$ instead.

1) $\Leftrightarrow$ Reflection changes sense of circulation of A- and B-sublattice terms in H_{SO}



But it also affects spin operators non-trivially:

$\vec{L} = \vec{r} \times \vec{p}$

$\Downarrow$

Under $x \to -x$

$L_x \to L_x$

$L_{y,z} \to -L_{y,z}$

$\Leftarrow$ this is valid for <u>all</u> angular momentum operators

$\Downarrow$

$S_x \to S_x$

$S_y \to -S_y$

$S_z \to -S_z$

$\Rightarrow$ thus, H_{SO} is invariant (arrows change directions and $\hat{S}^z \to -\hat{S}^z$)

2) Time-reversal : antiunitary operator $\begin{cases} i \to -i \\ \vec{S} \to -\vec{S} \end{cases}$

H_{SO} remains invariant

3) Translational and rotational symmetries are also respected.

$$c_{A/B\,\vec{r}\sigma} = \frac{1}{\sqrt{N_A}} \sum_{k\in BZ} e^{i\vec{k}\cdot\vec{r}}\, c_{A/B\,\vec{k}\sigma}$$

$$\sum_{r\in A} \sum_{j=1}^{3} \left(i\lambda\; c^{+}_{A\,\vec{r}+\vec{u}_j}\, \hat{S}^z\, c_{A\,\vec{r}} + h.c. \right)$$

$$= i\lambda \sum_{j=1}^{3} \sum_{r\in A} \frac{1}{N_A} \sum_{kk'} \Big(e^{-i\vec{k}(\vec{r}+\vec{u}_j)}\, c^{+}_{Ak}\, \hat{S}^z\, e^{i\vec{k}'\cdot\vec{r}}\, c_{Ak'}$$

$$- e^{-i\vec{k}\cdot\vec{r}}\, c^{+}_{Ak}\, \hat{S}^z\, e^{i\vec{k}(\vec{r}+\vec{u}_j)}\, c_{Ak'} \Big) = \Big(\underbrace{\frac{1}{N_A}\sum_{r\in A} e^{i\vec{r}(\vec{k}-\vec{k}')}}_{=\delta_{kk'}} \Big)$$

$$= i\lambda \sum_{k} \sum_{j=1}^{3} \left(e^{-i\vec{k}\cdot\vec{u}_j}\, c^{+}_{Ak}\, \hat{S}^z\, c_{Ak} - e^{i\vec{k}\cdot\vec{u}_j}\, c^{+}_{Ak}\, \hat{S}^z\, c_{Ak} \right)$$

$$= \sum_{j=1}^{3} \sum_{k\in BZ} 2\lambda\, \sin(\vec{k}\cdot\vec{u}_j)\; c^{+}_{Ak}\, \hat{S}^z\, c_{Ak}$$

$$\Lambda(\vec{k}) = 2\lambda \sum_{j=1}^{3} \sin(\vec{k}\vec{u}_j) = 2\lambda \Big[\sin(k_x a) + \sin\left(-\frac{k_x a}{2} + \frac{\sqrt{3}}{2} k_y a\right)$$

$$+ \sin\left(-\frac{k_x a}{2} - \frac{\sqrt{3}\, k_y a}{2}\right) \Big] = (2\lambda) \left[\sin(k_x a) - 2 \sin\left(\frac{k_x a}{2}\right) \cos\left(\frac{\sqrt{3}\, k_y a}{2}\right) \right].$$

$$\hat{H}_{so} = \sum_{k\in BZ} \Lambda(\vec{k}) \left(c^{+}_{Ak}\, \hat{S}^z\, c_{Ak} - c^{+}_{Bk}\, \hat{S}^z\, c_{Bk} \right)$$

$$= \sum_{k\in BZ} \Lambda(\vec{k})\; c^{+}_{k}\, \sigma^z\, \hat{S}^z\, c_k$$

acts in sublattice (A/B) basis

Note: $c^{+}_{Ak}\, \hat{S}^z\, c_{Ak} = \left(c^{+}_{Ak\uparrow}, c^{+}_{Ak\downarrow} \right) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c_{Ak\uparrow} \\ c_{Ak\downarrow} \end{pmatrix} =$

$$= \left(c^{+}_{Ak\uparrow} c_{Ak\uparrow} - c^{+}_{Ak\downarrow} c_{Ak\downarrow} \right).$$

Near K-point: $\vec{K} = \left(\frac{4\pi}{3a}, 0\right)$

$$\Lambda(\vec{K}) = 2\lambda\left[\underbrace{\sin\left(\frac{4\pi}{3}\right)}_{-\frac{\sqrt{3}}{2}} - 2\underbrace{\sin\left(\frac{1}{2}\,\frac{4\pi}{3}\right)}_{\frac{\sqrt{3}}{2}}\cos(0)\right] =$$

$$= 2\lambda\left(-\frac{\sqrt{3}}{2} - 2\,\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}\,\lambda$$

Near $\vec{K}' = -\vec{K}$: $\Lambda(\vec{K}') = 3\sqrt{3}\,\lambda$

Look on spin ↑ Hamiltonian:

$$H_\uparrow = \sum_{k\in BZ} \left(c^+_{Ak\uparrow}, c^+_{Bk\uparrow}\right)\begin{pmatrix}\Lambda(k) & \xi(k)\\ \xi^*(k) & -\Lambda(k)\end{pmatrix}\begin{pmatrix}c_{Ak\uparrow}\\ c_{Bk\uparrow}\end{pmatrix}$$

$$\det\begin{pmatrix}\Lambda - E & \xi\\ \xi^* & -\Lambda - E\end{pmatrix} = (\Lambda - E)(-\Lambda - E) - |\xi|^2 = 0$$

$$E^2 - \Lambda^2 - |\xi|^2 = 0$$

$$E = \pm\sqrt{\Lambda^2(k) + |\xi(k)|^2}$$

Look on spin-↓:

$$H_\downarrow = \sum_{k\in BZ} \left(c^+_{Ak\downarrow}, c^+_{Bk\downarrow}\right)\begin{pmatrix}-\Lambda(k) & \xi(k)\\ \xi^*(k) & \Lambda(k)\end{pmatrix}\begin{pmatrix}c_{Ak\downarrow}\\ c_{Bk\downarrow}\end{pmatrix}$$

$$\Rightarrow \det\begin{pmatrix}-\Lambda - E & \xi\\ \xi^* & \Lambda - E\end{pmatrix} = 0 \Rightarrow$$

$$E = \pm\sqrt{\Lambda^2(k) + |\xi|^2}$$

$\Rightarrow$ we can expect the gap at $\vec{K}, \vec{K}'$ in the full Hamiltonian $H = H_\uparrow + H_\downarrow$ (check later)

Look now in the vicinity of $\vec{K}$:

$$H_K = \sum_{\vec{k}\approx\vec{K}} \left(\psi^\dagger_{AKS}, \psi^\dagger_{BKS}\right) \hbar v\left(q_x\sigma^x + q_y\sigma^y\right)\begin{pmatrix}\psi_{AKS}\\ \psi_{BKS}\end{pmatrix}$$

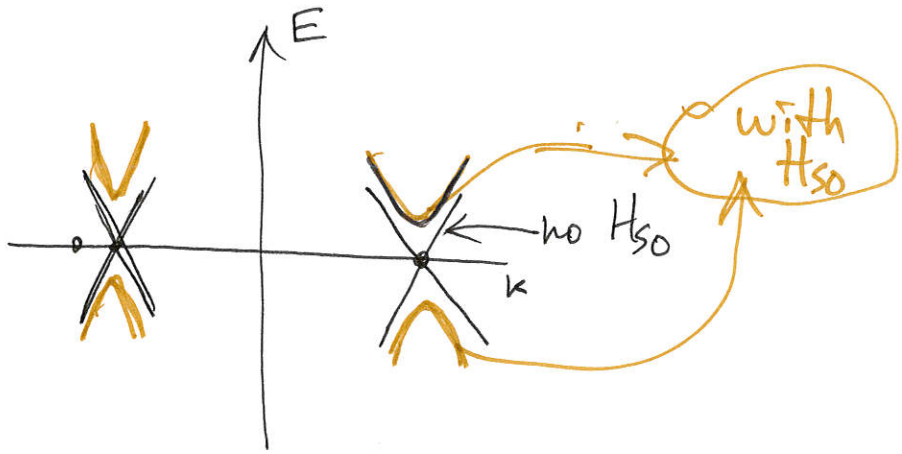
$(\vec{k}=\vec{K}+\vec{q})$

$$+ \underbrace{\Lambda(\vec{k}=\vec{K})}_{-3\sqrt{3}\lambda} \left(\psi^\dagger_{AKS}, \psi^\dagger_{BKS}\right) \sigma^z \hat{S}^z \begin{pmatrix}\psi_{AKS}\\ \psi_{BKS}\end{pmatrix} =$$

$$= \sum_{\vec{k}\approx\vec{K}} \left[\psi^\dagger_K(q)\ \hbar v\left(q_x\sigma^x + q_y\sigma^y\right) - m_{SO}\,\sigma^z\hat{S}^z\right]\psi_K(q)$$

Near $\vec{K}'$:

$$H_{K'} \approx \sum_{\vec{k}\approx\vec{K}'} \psi^\dagger_{K'}(\bar{q})\left(\hbar v\left(-q_x\sigma^x + q_y\sigma^y\right) + m_{SO}\,\sigma^z\hat{S}^z\right)\psi_{K'}$$



Full Hamiltonian

$$\begin{pmatrix} c^\dagger_{Ak\uparrow} & c^\dagger_{Bk\uparrow} & c^\dagger_{Ak\downarrow} & c^\dagger_{Bk\downarrow} \end{pmatrix} \begin{pmatrix} \Lambda & \xi & 0 & 0 \\ \xi^* & -\Lambda & 0 & 0 \\ 0 & 0 & -\Lambda & \xi \\ 0 & 0 & \xi^* & \Lambda \end{pmatrix} \begin{pmatrix} c_{Ak\uparrow} \\ c_{Bk\uparrow} \\ c_{Ak\downarrow} \\ c_{Bk\downarrow} \end{pmatrix}$$

$$\det \begin{pmatrix} \Lambda - E & \xi & 0 & 0 \\ \xi^* & -\Lambda - E & 0 & 0 \\ 0 & 0 & -\Lambda - E & \xi \\ 0 & 0 & \xi^* & \Lambda - E \end{pmatrix} = 0$$

$$(\Lambda - E) \begin{vmatrix} -\Lambda - E & 0 & 0 \\ 0 & -\Lambda - E & \xi \\ 0 & \xi^* & \Lambda - E \end{vmatrix}$$

$$-\xi \begin{vmatrix} \xi^* & 0 & 0 \\ 0 & -\Lambda - E & \xi \\ 0 & \xi^* & \Lambda - E \end{vmatrix} = (\Lambda - E)(-\Lambda - E)\left[(-\Lambda - E)(\Lambda - E) - |\xi|^2\right]$$

$$-\xi\, \xi^* \left[(-\Lambda - E)(\Lambda - E) - |\xi|^2\right] =$$

$$= (E + \Lambda)(E - \Lambda)\left((E + \Lambda)(E - \Lambda) - |\xi|^2\right)$$

$$- |\xi|^2 \left((E + \Lambda)(E - \Lambda) - |\xi|^2\right)$$

$$= \left(E^2 - \Lambda^2 - |\xi|^2\right)^2 \Rightarrow E_\pm(k) = \pm\sqrt{\Lambda_k^2 + |\xi_k|^2}$$

has the same spectrum.

Write $H = \sum_k c^+_{ks}\, \vec{h}(k)\cdot\vec{\sigma}\, c_{ks}$

Expressions on page 150 show that

$$\vec{h}(\vec{k}=\vec{K}+\vec{q}) = \hbar v_F (q_x\hat{x}+q_y\hat{y}) - m_{so}\,\hat{s}^z\,\hat{z} =$$

$$= \hbar v_F (q_x\hat{x}+q_y\hat{y}) \mp m_{so}\,\hat{z} \quad \left(\begin{matrix}\text{upper sign } \uparrow \\ \text{lower } \downarrow\end{matrix}\right)$$

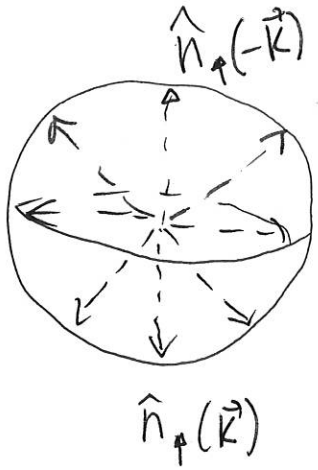
$$\vec{h}(\vec{k}=-\vec{K}+\vec{q}) = \hbar v_F (-q_x\hat{x}+q_y\hat{y}) + m_{so}\,\hat{s}^z\,\hat{z} =$$

$$= \hbar v_F (-q_x\hat{x}+q_y\hat{y}) \pm m_{so}\,\hat{z}$$

Construct unit vector $\hat{n}_s(K) = \dfrac{\vec{h}(\vec{k}+\vec{q})}{|\vec{h}(\vec{k}+\vec{q})|}$ for spin-s

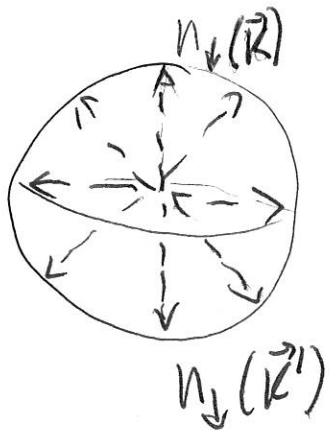
$$\hat{n}_s(K') = \frac{\vec{h}(k'+q)}{|\vec{h}(k'+\vec{q})|} \quad \text{for spin-s}$$

$s = \uparrow$



topologically non-trivial bands.

$s = \downarrow$



Observe that

$\hat{n}_\uparrow(\vec{K})$ cover <u>lower</u> hemisphere while $\hat{n}_\uparrow(\vec{K}')$ covers <u>upper</u> hemisphere

$\Rightarrow$ spin-$\uparrow$ covers the whole sphere!

Similarly

$\hat{n}_\downarrow(\vec{K})$ covers <u>upper</u> half sphere

$\hat{n}_\downarrow(\vec{K}')$ covers <u>lower</u>

$\Rightarrow$ spin-$\downarrow$ covers the whole but in opposite to $s=\uparrow$ sense.

Thus, we have two copies of topological insulator with opposite Q's for spin-$\uparrow$ and spin-$\downarrow$.
Crucial for this is the fact that band gaps at K and K' for spin-s have different signs ("minus" at K, "plus" at K' } $\uparrow$
"+" at K, "−" K' } $\downarrow$)

For the case of usual insulator which has gap due to asymmetry between A and B sublattices (problem 2 in HW4), the gap m has the same sign at K and K', and as a result unit vector $\hat{n}$ does not cover the full sphere (only upper hemisphere.)

Edge states (Kane, Mele PRL 95 226801 (2005))

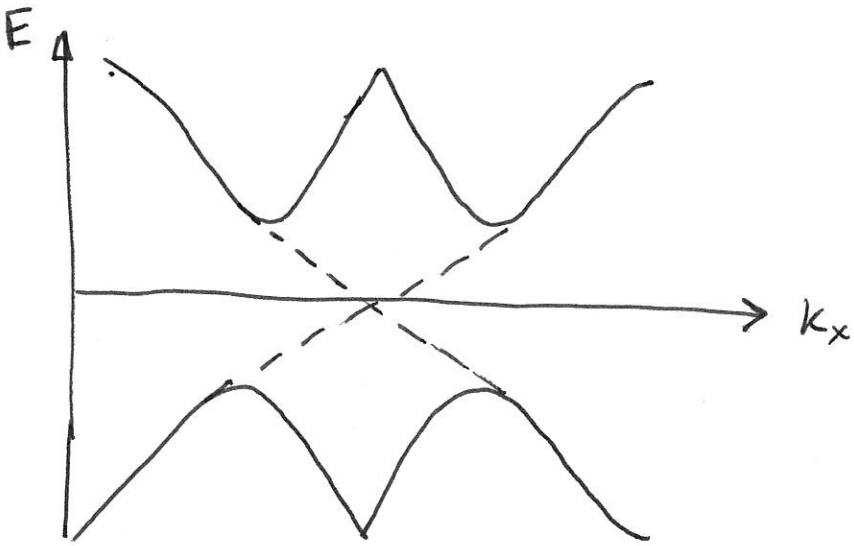


Fig. 1
Zig-zag edge

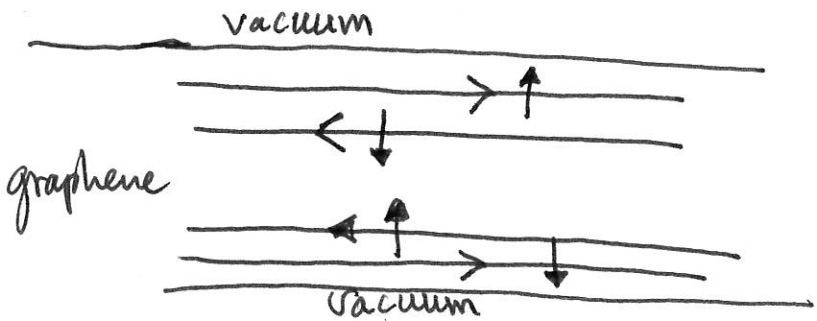


Fig. 2(a)
spin-filtered edges: $\uparrow$ and $\downarrow$ move in opposite directions

Properties of edge states: - localized at distance $\sim m_{so}^{-1}$ from the edge

- group velocity $\vec{v} = \frac{1}{\hbar} \frac{\partial \hat{E}(\vec{k})}{\partial \vec{k}}$ on the edge

is positive for $s = \uparrow$ and negative for $s = \downarrow$:

edge states are spin-filtered. (opposite on the other edge)

$\left.\begin{matrix} s \to -s \\ v \to -v \end{matrix}\right\}$ are related by time-reversal symmetry!

What is $\hat{\Theta}$ (time reversal)? (see Rev. Mod. Phys 82, 3045 (2010) by Hasan and Kane)

$$\hat{\Theta} = e^{i\pi \hat{S}_y/\hbar}\, \hat{K}$$

$\hat{S}_y$ = spin op. $\hat{K}$ ← complex conjugation operator

For $s = 1/2$, $e^{i\pi \hat{S}_y/\hbar} = i\sigma_y \Rightarrow \underline{\hat{\Theta} = i\sigma_y K}$ for $s = 1/2$ spins

$\hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y$

$$i\sigma_y = i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \longrightarrow$$

then $\hat{\Theta}^2 = (i\sigma_y K)(i\sigma_y K) =$

$$= \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix} K \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix} K =$$

$$= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} K^2 =$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = (-1)\,\sigma_0$$

$\Rightarrow \hat{\Theta}^2 = -1$ for $s = 1/2$.

<u>Kramer's theorem</u>: in $\hat{\Theta}$-invariant systems states are 2-fold degenerate

that is, let $[\hat{H}, \hat{\Theta}] = 0$

$$\hat{H}|\psi\rangle = E|\psi\rangle \Rightarrow \underbrace{\hat{\Theta}\hat{H}|\psi\rangle} = \hat{H}(\hat{\Theta}|\psi\rangle)$$

$$\hat{\Theta} E|\psi\rangle = E(\hat{\Theta}|\psi\rangle)$$

$\Rightarrow$ if $|\psi\rangle$ is an eigenstate with energy E, then $\hat{\Theta}|\psi\rangle$ is an eigenstate with E too.

<u>Important</u>: $\hat{\Theta}|\psi\rangle$ and $|\psi\rangle$ are distinct states

proof by contradiction: suppose the opposite,

then $\hat{\Theta}|\psi\rangle = c|\psi\rangle$

$\hookrightarrow$ some complex coefficient

$$\Rightarrow \hat{\Theta}^2|\psi\rangle = \hat{\Theta}\, c|\psi\rangle = c^* \hat{\Theta}|\psi\rangle =$$

$$= c^*(c|\psi\rangle) = |c|^2|\psi\rangle$$

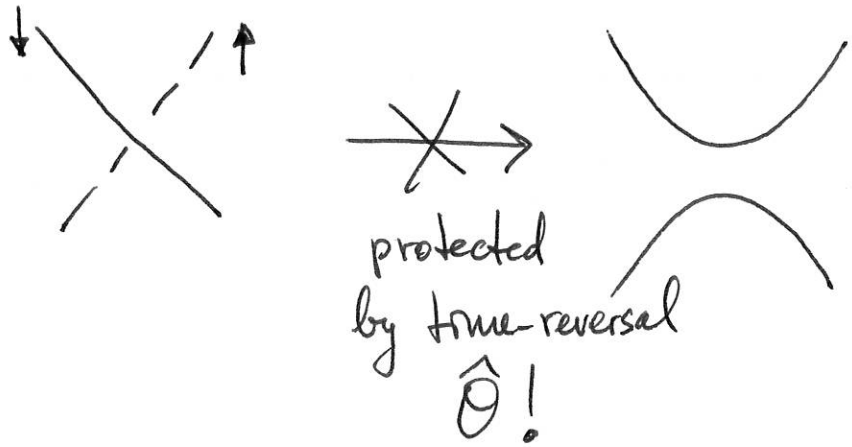
but $\hat{\Theta}^2 = -1 \Rightarrow \hat{\Theta}^2|\psi\rangle = -|\psi\rangle$

$\Rightarrow |c|^2 = -1$ which is contradiction.

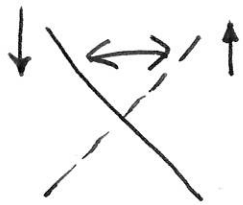
$\Rightarrow$ $\hat{\Theta}|\psi\rangle$ is distinct from $|\psi\rangle$: $\langle\psi|\hat{\Theta}|\psi\rangle = 0$.

(simple example: $|\psi\rangle = |\uparrow\rangle \Rightarrow \hat{\Theta}|\psi\rangle = |\downarrow\rangle$)

⇒ back to edge states:



Simple reason: impurity scattering must change not only direction of propagation, but also spin — not possible if spin is conserved.



only magnetic impurities can do that but they break $\hat{\Theta}$!

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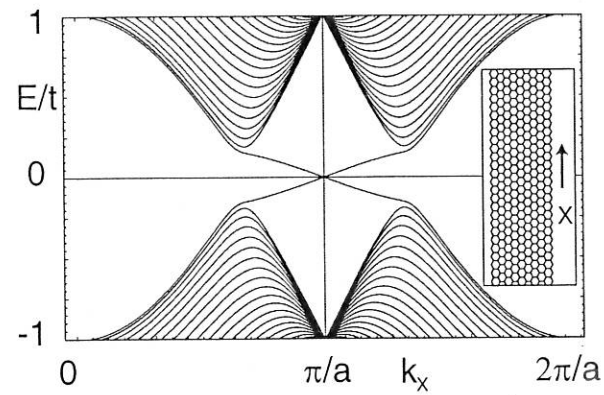
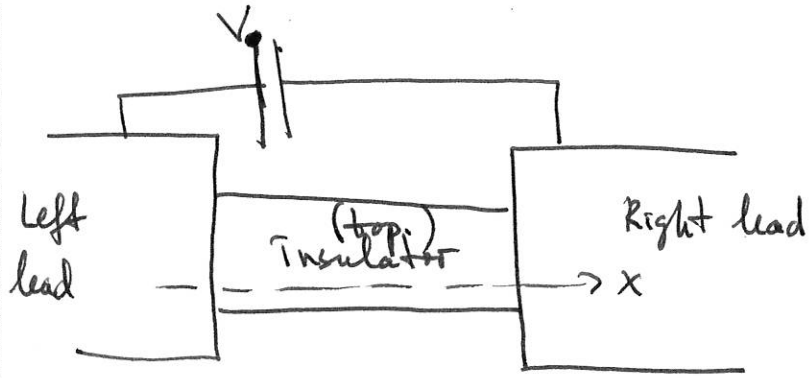


FIG. 1. (a) One-dimensional energy bands for a strip of graphene (shown in inset) modeled by (7) with $t_2/t = 0.03$. The bands crossing the gap are spin filtered edge states.

Conductance experiments (Molenkamp's group, Science 318 p. 766 (2007))



Conductance $G = \frac{I}{V}$

V raises chem. potential on the left by eV relative to that on the right ⇒ there are more electrons coming from the left ⇒ current flows

$$I = \underbrace{e\,\delta n}_{\text{density of excess electrons}}\; v_F\; T(E_F)$$

← transmission probability from left to right

← their velocity

$$v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m}\frac{\sqrt{2mE_F}}{\hbar} = \sqrt{\frac{2E_F}{m}}$$

for spinless electrons →

$$\delta n = \int_{\sqrt{2m(E_F+eV)}/\hbar} \frac{dk}{2\pi} = \frac{\sqrt{2m}}{2\pi\hbar}\left(\sqrt{E_F+eV} - \sqrt{E_F}\right) =$$

$$\frac{\sqrt{2mE_F}}{\hbar} \qquad = \frac{\sqrt{2m}}{2\pi\hbar}\,\frac{1}{2}\,\frac{eV}{\sqrt{E_F}} = \sqrt{\frac{m}{2E_F}}\left(\frac{eV}{2\pi\hbar}\right)$$

$$\Rightarrow I = e\sqrt{\frac{m}{2E_F}}\;\frac{eV}{h}\sqrt{\frac{2E_F}{m}}\;T(E_F) = \left(\frac{e^2}{h}\right)T(E_F)\,V$$

$$\frac{e^2}{h} \cong \frac{1}{25\,k\Omega} = \text{conductance quantum}$$

$$G = \frac{I}{V} = \frac{e^2}{h} T(E_F)$$

$\Rightarrow$ In case of <u>perfect</u> transmission $G = \frac{e^2}{h}$

(finite quantized G due to leads only!)

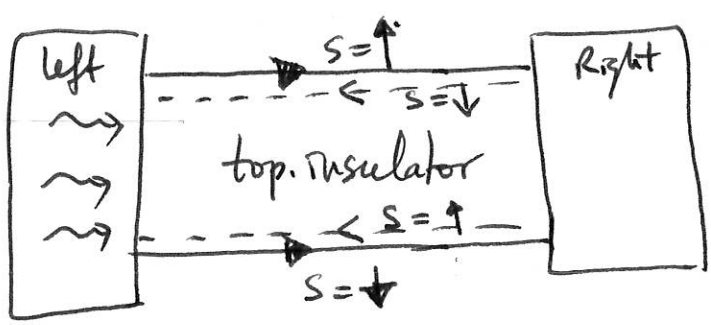
If several conducting channels (due to spin, for example)

$$G = \frac{e^2}{h} \sum_n T_n(E_F) .$$

$\Rightarrow$ $G \sim e^{-gap/T}$ for usual insulator

$G = 2\frac{e^2}{h}$ for topological insulator

(even in presence of non-magnetic disorder, temperature T, etc.)



dashed trajectories do not carry current — states on the left are occupied (below $E_F + eV$).

Solid trajectories carry spin-resolved current from left to right.

HgTe Science 318 766 (2007)

Although the four-band Dirac model (Eq. 1) gives a simple qualitative understanding of this novel phase transition, we also performed more realistic and self-consistent eight-band **k·p** model calculations (*13*) for a 6.5-nm quantum well, with the fan chart of the Landau levels displayed in Fig. 1B. The two anomalous Landau levels cross at a critical magnetic field $B^c_\perp$, which evidently depends on well width. This implies that when a sample has its Fermi energy in the gap at zero magnetic field, this energy will always be crossed by the two anomalous Landau levels, resulting in a QH plateau in-between the two crossing fields. Figure 3 summarizes the dependence of $B^c_\perp$ on well width d. The open red squares are experimental data points that result from fitting the eight-band **k·p** model to experimental data as in Fig. 1, while the filled red triangles result solely from the **k·p** calculation. For reference, the calculated gap energies are also plotted in this graph as open blue circles. The band inversion is reflected in the sign change of the gap. For relatively wide wells (d > 8.5 nm), the (inverted) gap starts to decrease in magnitude. This is because for these well widths, the band gap no longer occurs between the $E1$ and $HH1$ levels, but rather between $HH1$ and $HH2$—the second confined hole-like level, as schematically shown in the inset of Fig. 3 [see also (*17*)]. Also in this regime, a band crossing of conductance- ($HH1$) and valence- ($HH2$) band–derived Landau levels occurs with increasing magnetic field (*13*, *17*, *18*). Figure 3 clearly illustrates the quantum phase transition that occurs as a function of d in the HgTe QWs: Only for $d > d_c$ does $B^c_\perp$ exist, and at the same time the energy gap is negative (i.e., the band structure is inverted). The experimental data allow for a quite accurate determination of the critical thickness, yielding d_c = 6.3 ± 0.1 nm.

Zero-field edge channels and the QSH effect. The actual existence of edge channels in insulating inverted QWs is only revealed when studying smaller Hall bars [the typical mobility of 10^5 cm^2 V^{-1} s^{-1} in n-type material implies an elastic mean free path of $l_{mfp} \approx$ 1 μm (*19*, *20*)—and one may anticipate lower mobilities in the nominally insulating regime]. The pertinent data are shown in Fig. 4, which plots the zero B-field four-terminal resistance $R_{14,23} \equiv V_{23}/I_{14}$ as a function of normalized gate voltage (V_{thr} is defined as the voltage for which the resistance is largest) for several devices that are representative of the large number of structures we investigated. $R_{14,23}$ is measured while the Fermi level in the device is scanned through the gap. In the low-resistance regions at positive $V_g - V_{thr}$, the sample is n-type; at negative $V_g - V_{thr}$, the sample is p-type.

The black curve labeled I in Fig. 4 was obtained from a medium-sized [(20.0 × 13.3) μm^2] device with a 5.5-nm QW and shows the behavior we observe for all devices with a normal band structure: When the Fermi level is in the gap, $R_{14,23}$ increases strongly and is at least several tens of megohm (this is the detection limit of the lock-in equipment used in the experiment). This clearly is the expected behavior for a conventional insulator. However, for all devices containing an inverted QW, the resistance in the insulating regime remains finite. $R_{14,23}$ plateaus at well below 100 kilohm (i.e., $G_{14,23}$ = 0.3 e^2/h) for the blue curve labeled II, which is again for a (20.0 × 13.3) μm^2 device fabricated by optical lithography, but that contains a 7.3-nm-wide QW. For much shorter samples (L = 1.0 μm, green and red curves III and IV) fabricated from the same wafer, $G_{14,23}$ actually reaches the predicted value close to $2e^2/h$, demonstrating the existence of the QSH insulator state for inverted HgTe QW structures.

Figure 4 includes data on two devices with d = 7.3 nm, L = 1.0 μm. The green trace (III) is from a device with W = 1.0 μm, and the red trace (IV) corresponds to a device with W = 0.5 μm. Clearly, the residual resistance of the devices does not depend on the width of the structure, which indicates that the transport occurs through edge channels (*21*). The traces for the d = 7.3 nm, L = 1.0 μm devices do not reach all the way into the p-region because the electron-beam lithography needed to fabricate the devices increases the intrinsic (V_g = 0 V) carrier concentration. In addition, fluctuations on the conductance plateaus in traces II, III, and IV are reproducible and do not stem from, e.g., electrical noise. Although all $R_{14,23}$ traces discussed so far were taken at the base temperature (30 mK) of our dilution refrigerator, the conductance plateaus are not limited to this very-low-temperature regime. In the inset of Fig. 4, we reproduce the green 30-mK trace III on a linear scale and compare it with a trace (in black) taken at 1.8 K from another ($L \times W$) = (1.0 × 1.0) μm^2 sample, which was fabricated from the same wafer. In the fabrication of this sample, we used a lower-illumination dose in the e-beam lithography, resulting in a better (but still not quite complete) coverage of the n-i-p transition. Clearly, in this further sample, and at 1.8 K, the $2e^2/h$ conductance

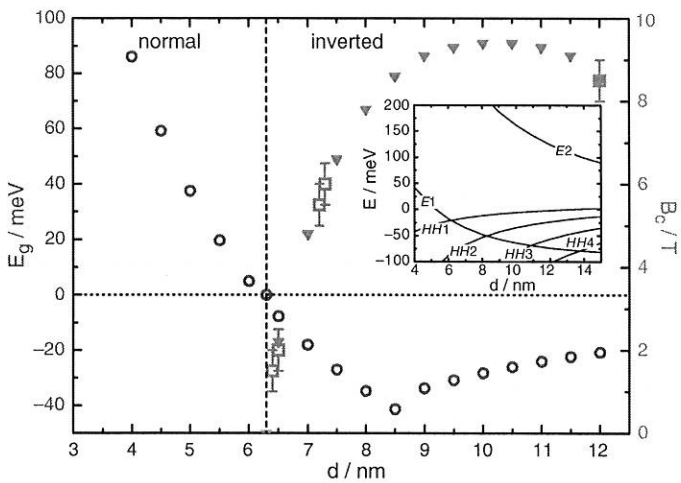


Fig. 3. Crossing field, $B^c_\perp$ (red triangles), and energy gap, E_g (blue open dots), as a function of QW width d resulting from an eight-band **k·p** calculation. For well widths larger than 6.3 nm, the QW is inverted and a mid-gap crossing of Landau levels deriving from the $HH1$ conductance and $E1$ valence band occurs at finite magnetic fields. The experimentally observed crossing points are indicated by open red squares. The inset shows the energetic ordering of the QW subband structure as a function of QW width d. [See also (*17*)].

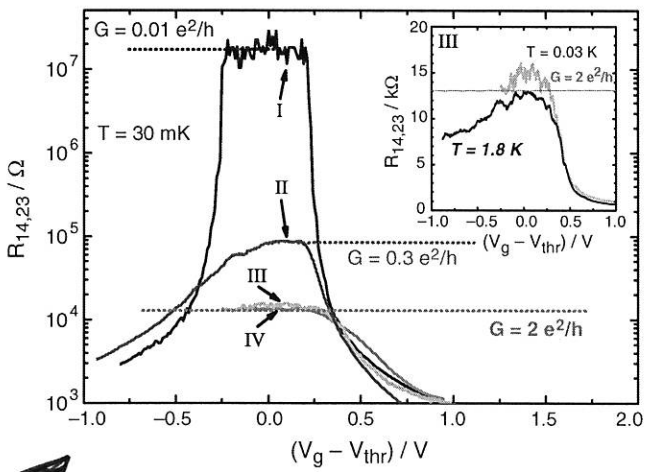


Fig. 4. The longitudinal four-terminal resistance, $R_{14,23}$, of various normal (d = 5.5 nm) (I) and inverted (d = 7.3 nm) (II, III, and IV) QW structures as a function of the gate voltage measured for B = 0 T at T = 30 mK. The device sizes are (20.0 × 13.3) μm^2 for devices I and II, (1.0 × 1.0) μm^2 for device III, and (1.0 × 0.5) μm^2 for device IV. The inset shows $R_{14,23}(V_g)$ of two samples from the same wafer, having the same device size (III) at 30 mK (green) and 1.8 K (black) on a linear scale.

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